

Sera Gunn

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EMPLOYMENT

Doctoral Lecturer College of Staten Island,
Aug 2023–present City University of New York (CUNY)
Research/Teaching Assistant Georgia Institute of Technology
Aug 2017–Aug 2023

EDUCATION

PhD Georgia Institute of Technology
Aug 2017–Aug 2023 Thesis: *Divisors and Multiplicities Under Tropical and Signed Shadows*
Advisor: Prof. Matthew Baker
Bachelor of Mathematics University of Waterloo
Sept 2013–June 2017 *Honours Pure Math & Combinatorics and Optimization*
(With Distinction – Dean's Honours List)

TECHNICAL SKILLS

▶ github.com/seragunn

I do a lot of documentation-type work. Mainly using Quarto/HTML/CSS.

- I'm the current lead maintainer for our department website <https://www.math.csi.cuny.edu> (▶ [github repository](#))
- My online notes for a [course in combinatorics](#) using Quarto/Sagemath/Some Javascript
- My lab activities for a [probability and statistics course](#) using Quarto/R in WebAssembly

RESEARCH AREA

My research was in the connections between combinatorics and algebraic geometry. In particular, tropical geometry, matroids, hyperfields and ordered blueprints. My PhD work involved studying polynomials over hyperfields/blueprints.

TEACHING AND OUTREACH

Course Instructor, College of Staten Island

Class sizes of about 30–40 students

Course #	Course Description	Times	Last taught
(650)	Disc. Math. Modeling for Sec. School Teachers	1	Spr. 2024
(337)	Applied Combinatorics & Graph Theory	1	Spr. 2024
(232)	Analytic Geometry and Calculus II	2	Fall 2025
(231)	Analytic Geometry and Calculus I	2	Spr. 2024
(229)	Calculus Computer Laboratory (with Julia)	3	Fall 2025
(125)	College Algebra and Trigonometry (recitation)	4	Fall 2025
(123)	College Algebra and Trigonometry	2	Fall 2024
(113)	Intro. to Probability and Statistics (w/Computer Lab)	2	Fall 2025

Course Instructor, Georgia Tech

Class sizes of about 50–80 students.

Course #	Course Description	Times
(3215)	Introduction to Probability and Statistics	1
(3012)	Applied Combinatorics	1

Teaching Assistant, Georgia Tech

Class sizes of about 20–30 students

Except for 3012, this involved teaching a weekly recitation class

Course #	Course Description	Times
(3012)	Applied Combinatorics	4
(2603)	Introduction to Discrete Mathematics	1
(2551)	Multivariable Calculus	1
(2550)	Introduction to Multivariable Calculus	1
(1564)	Linear Algebra with Abstract Vector Spaces	1
(1554)	Linear Algebra	1
(1553)	Introduction to Linear Algebra	2
(1552)	Integral Calculus	1
(1803)	Precalculus	1
(1711)	Finite Mathematics	1

High School Math Day, Georgia Tech

When I started, this was called the High School Math Competition (HSMC) and consisted of a yearly event where students from high schools in the Atlanta area were invited to Georgia Tech for a math competition event. The event consisted of several parts: individual, rapid questions, and either a group or proof based competition.

In my first 3 years at Georgia Tech I was involved as a volunteer. Then during the COVID-19 pandemic (Fall 2020 to Summer 2021) I was the solo organizer for the event. During my last year, I was joined by two visiting assistant professors/postdoctoral researchers in organizing. It was this last year where we worked on shifting the HSMC event to the HSMD (High School Math Day) event.

The shift from "Math Competition" to "Math Day" represented a push to make the event more attractive and accessible to non-STEM students and underrepresented groups. In addition to the responsibilities associated to the competition, we are also in the process of redesigning the event from the ground up and advertising to a broader selection of schools in the Atlanta area.

Directed Reading Mentor, Georgia Tech

Fall 2019–Fall 2022

The Directed Reading Program consists of weekly 1 hour meetings with one or two students (per semester) to guide them through extracurricular mathematics material. The students are expected to complete about 3 to 4 hours of reading outside of the weekly meetings. Projects I have mentored on include: algebraic number theory (over two semesters), additional topics in combinatorics, and an introduction to ordinary and exponential generating functions.

PROFESSIONAL SERVICE

College of Staten Island

Fall 2024–current	representative	Undergraduate Curriculum Committee
Fall 2024–current	member	Pre-major committee
Fall 2024–current	lead maintainer	Math Department Webpage
2024–2025	member	Math Department Hiring Committee

Georgia Tech

2019–2023	co-organizer	Georgia Tech Algebraic Student Seminar
2022–2023	co-organizer	Graduate Student Colloquium
2022–2023	member	Graduate Student Council
2022–2023	co-organizer	Georgia Tech High School Math Day
2021–2022	lead organizer	Georgia Tech High School Math Competition
2017–2021	volunteer	Georgia Tech High School Math Competition
Nov. 2021	co-organizer	Tropical Geometry Seminar
Fall 2019	co-organizer	Learning Seminar on Lorentzian Polynomials
2018–2019	co-organizer	Georgia Tech Research Horizons Seminar

CONFERENCE TALKS

August 2022	<i>Factoring Polynomials over Hyperfields</i>	Universität Regensburg
August 2019	<i>Construction of Fully Faithful Tropicalizations for Curves</i>	Universität Regensburg

PUBLICATIONS

- [1] Sera Gunn. “Tropical extensions and Baker-Lorscheid multiplicities for idylls”. In: *Communications in Algebra* 53.1 (2025), pp. 63–89. doi: [10.1080/00927872.2024.2372376](https://doi.org/10.1080/00927872.2024.2372376).
- [2] Sera Gunn and Philipp Jell. “Construction of Fully Faithful Tropicalizations for Curves in Ambient Dimension 3”. In: *Beiträge zur Algebra und Geometrie* (Dec. 2024). doi: [10.1007/s13366-024-00780-9](https://doi.org/10.1007/s13366-024-00780-9).
- [3] Andreas Gross and Sera Gunn. *Factoring multivariate polynomials over hyperfields and the multivariable Descartes’ problem*. July 2023. arXiv: [2307.09400](https://arxiv.org/abs/2307.09400) [[math.AG](https://arxiv.org/archive/math)].

CONFERENCES AND WORKSHOPS ATTENDED

2022	Aug 1–5	Young Researchers’ Conference on Nonarchimedean and Tropical Geometry	Universität Regensburg
	Apr 1–3, 2022	Georgia Algebraic Geometry Symposium	Emory University
2021	Nov 11 & 18	Tropical Geometry Seminar	Georgia Tech
	Apr 12–16	Algebraic Geometry and Polyhedra	ICERM (virtual)
2020	Mar 7–11	Arizona Winter School (Non-Abelian Chabauty)	University of Arizona
2019	Fall (Weekly)	Learning Seminar on Lorentzian Polynomials	Georgia Tech
	Fall (Weekly)	Learning Seminar on Non-Abelian Chabauty	Emory University
	Jul 29–Aug 2	Young Researchers’ Conference on Nonarchimedean and Tropical Geometry	Universität Regensburg
	Apr 13 & 14	Meeting on Applied Algebraic Geometry	Georgia Tech
	Mar 30–31	GTTACA: Georgia Tech Tropical, Arithmetic and Combinatorial Algebraic-geometry	Georgia Tech
	Mar 2–6	Arizona Winter School (Topology and Arithmetic)	University of Arizona
2018	June 28–29	Summer School on Hyperbolic Polynomials, Sums of Squares, and Optimization	Georgia Tech

Mar. 2–6	Arizona Winter School (Iwasawa Theory)	University of Arizona
Feb 23–25	Georgia Algebraic Geometry Symposium	Georgia Tech

SCHOLARSHIPS & AWARDS

- 2022 Outstanding TA Award
- 2020 FESTA Fellowship
- 2020 Outstanding Math Lab Tutor
- 2017 NSERC Undergraduate Student Research Award
Topic: *Tropical scheme theory*
- 2016 NSERC Undergraduate Student Research Award
Topic: *Enumerating maps via symmetric functions*
- 2013 University of Waterloo President's Scholarship